EFY TESTED S.C. Dwivedi

Control Four Appliances With A Touch Switch Panel

Ashok Baijal

This project is based on a capacitive touch sensor for controlling four electrical appliances. Toggle switches have been used for operating lights or fans, and momentary press switches for operating electric doorbells. Most of these switches are mechanical and susceptible to damage through dirt, moisture or mishandling.

Mechanical switches are being replaced by touch switches in some modern homes. Touch switches do not have mechanical wear-and-tear problems and are normally enclosed in suitable boxes, making these resilient in the toughest environments. Some touch switches are designed beautifully, making them aesthetically-pleasing and blend with the room decor. But, commercially-available touch switches are costly. The circuit presented here is low-cost and easy to build.

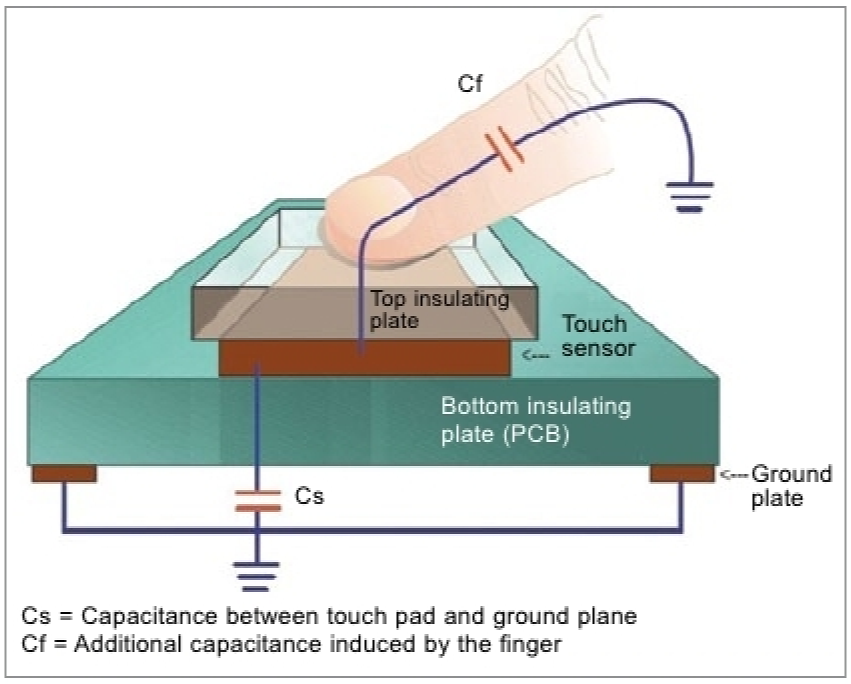

Fig. 1: Working principle of a touch switch

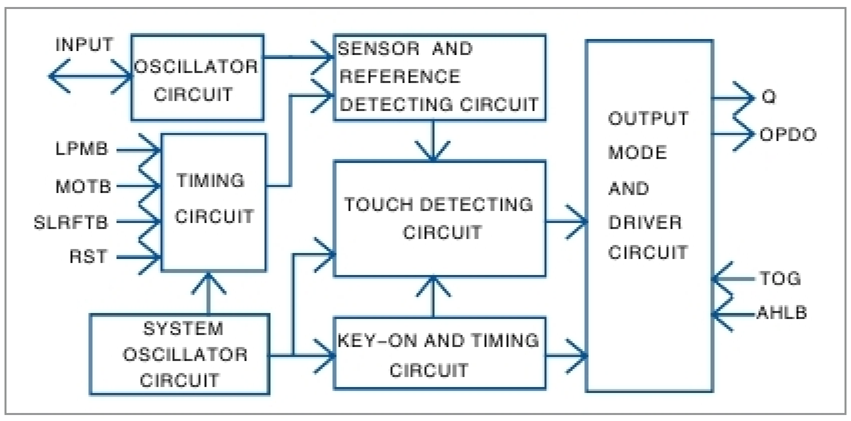

Fig. 2: Block diagram of TTP223 IC

How the touch switch works

A capacitive sensor works by detecting a change in capacitance due to influence of an external object. The touch plate is covered with insulating material, and the user does not come in contact with the electrical circuit.

A capacitive touch switch has different layers—top insulating face plate followed by touch plate, another insulating layer and then ground plate, as shown in Fig. 1.

In practice, a capacitive sensor can be made on a double-sided PCB by regarding one side as the touch sensor and the opposite side as ground plate of the capacitor. When power is applied across these plates, the two plates get charged. In equilibrium state, the plates have the same voltage as the power source.

Capacitance is the measure of charge per volt (Cs = Q/V). When a finger is moved near the sensor plate, electric field around the capacitor plate polarises the human body and energy gets transferred to it. This causes more charge to flow to the sensor plate till another equilibrium state is reached. This increases the charge concentration on the sensor plate, causing a change (increase) in capacitance.

It is not necessary to physically touch the sensor plate as disturbance of the electric field near the sensor is enough to induce a change in capacitance. The sensor plate can therefore be placed behind a protective overlay. Induced capacitance is determined by the following relationship:

$$Cf = k\,\varepsilon_0(A/d)$$

where, k is dielectric constant of the protective overlay, ε_0 is permittivity of vacuum, A is area of the touch plate and d is separation distance between the conductive plate and the finger.

Dielectric constant of glass ranges between 3.8 and 14.5, while that of plexiglass (acrylic) is between 2.6 and 3.5, of paper is 3.6, and wood has a dielectric constant between 1.4 and 2.9. As such, glass is the best material for the protective overlay, followed by plexiglass.

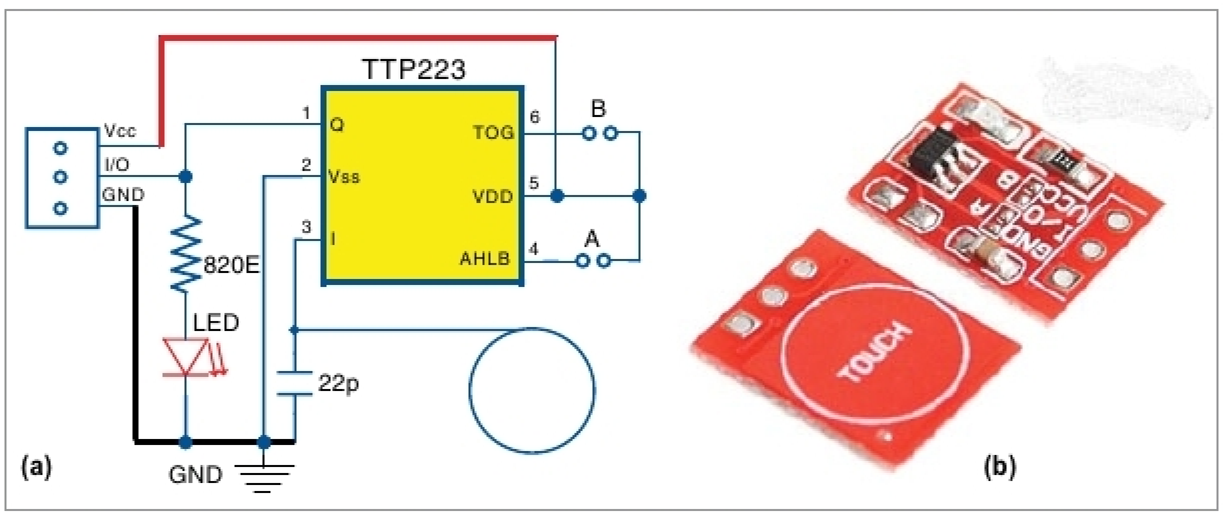

Fig. 3: (a) Circuit diagram of TTP223 module; (b) PCB of TTP223 module